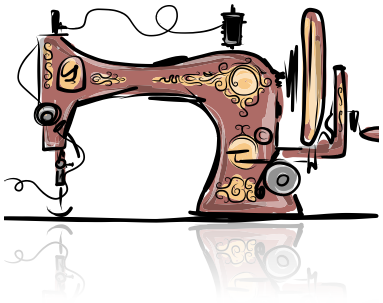



Linear Motion

**Multiple motion in a sewing machine**



- Motion of the needle

  _____

- Motion of the wheel

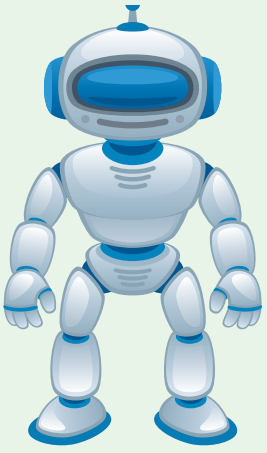
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- Motion of footrest

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## 2.3 Science Today - Robot

Robots are automatic machines. Some robots can perform mechanical and repetitive jobs faster and more accurately than people. Robots can also handle dangerous materials and explore distant planets.



A bipedal humanoid robot.

The term 'robot' comes from a czech word, 'robota' meaning 'forced labour'. Robotics is the science and study of robots.

### What can Robots do?

Robots can sense and respond to their surroundings. They can handle delicate objects or apply great force. For example, they can perform eye operations guided by a human surgeon, or assemble a car. With **artificial intelligence**, robots will also be able to make decisions for themselves.

### How do Robots sense?

Electronic sensors function as robot's eyes and ears. Twin video cameras give the robot a 3-D view of the world. Microphones detect sounds. Pressure



The quadrupedal military robot